Seclusion EDA Findings – CRDC 2021-22

Prepared for: Adhya Putta · October Milestone · Scope: 2021-22 school-level data, with 2017-18 vs 2021-22 comparison

Bottom line

Seclusion in 2021-22 is still rare (7.1% of schools report any), heavily skewed, and driven by state reporting context. The same target definition used for 2015-16 and 2017-18 works here: a **binary any-seclusion target** on schools with at least 30 students, an **elevated tier** as a sensitivity target, and models run **with and without state**. Two things are new this year: **virtual schools have no outcome at all**, and **Section 504 and English-learner counts are heavily missing**. The 2020-21 COVID check and race/sex equity could not be done because those data were not supplied.

Data completeness

- 98,010 schools; 93,094 (95.0%) are analysis-ready. The cleaning log reconciles exactly with the missing values in the schools file.
- Seclusion is missing for 4,905 schools (5.0%). **All 2,496 virtual schools are among them** (51% of the missing, all coded -9), so virtual vs non-virtual seclusion cannot be compared. The other 2,409 are structural skips (1,881), suppressed (353) or missing (175).
- Dropped schools are smaller (median 258 students vs 412) and more often special-education (5.6% vs 1.6%).
- Section 504 counts are 22.6% missing and English-learner counts 18.8%, concentrated in a few states. The gaps are informative (English-learner-missing schools report 4.8% vs 7.5%), so keep the rows and add missing indicators.
- The nonbinary count is 97.6% missing, so the four partial-enrollment flags fire for about 97% of schools and carry no information.

What the data show

- **Distribution:** 92.9% of schools report zero. Among the 6,600 that report any, the median is 4 instances, the 90th percentile 36, the maximum 2,058. The top 100 schools hold 27.8% of instances.

- **State dominates:** Wisconsin, Connecticut, Iowa, Maine, Washington and Kansas have 22-34% of schools reporting any; Florida, Georgia, Texas, California and New York are near or under 1.5%; Hawaii and Puerto Rico report none. State alone gives ROC-AUC 0.80, and adding it lifts a baseline model's average precision from 0.17 to 0.35.
- **Concentration:** 13.6% of districts have any school reporting seclusion; 110 districts hold half of all instances. One AZ school reports 2,058 instances for 11 students.
- **School type:** special-education schools 17.2% any (vs 6.9%); justice facilities 4.6%; alternative 1.8%, charter 2.4%, magnet 4.3%. Elementary 9.9%, middle 5.5%, high 3.4%. 94.7% of positive schools are ordinary schools.
- **Population:** %IDEA is the clearest positive correlate (2.9% any under 5% IDEA, about 8.6% at 15-30%). %504 is the strongest negative (8.9% at 0.5% or less, 2.3% at 10% or more). All rank correlations are weak (|rho| ⩽ 0.07).
- **Staffing:** non-monotonic and confounded. After adjustment, psychologists and social workers are positive; security guards, nurses and law-enforcement staff negative; counselors about zero. 641 schools fail simple FTE plausibility rules, while the cleaning outlier flag catches only 1.
- **Small schools distort rates:** schools under 30 students are 3.5% of schools but 41% of those with more than 100 instances per 100 students.
- **Equity (disability only):** students with IDEA are 13.6% of students but 79.1% of instances: 0.99 per 100 vs 0.040 for students without a disability (about 25x). These are events, not unique students.

2017-18 vs 2021-22

- 83,527 schools appear in both. Prevalence rose from 6.79% to 7.26% in the matched set, while total instances fell 27% (129,686 to 94,738): more schools report some seclusion, but fewer instances overall.
- Persistence is high: 43.0% of schools with any in 2017-18 also have any in 2021-22, versus 4.6% of the others. 36.1% of elevated schools stay elevated.
- State prevalence is nearly identical across years (r = 0.97). The biggest rises are Connecticut, South Dakota, Washington, Wyoming and Nebraska; the biggest falls Maryland, DC, Alabama and Florida.
- Across three waves (80,241 schools valid in all): 5.73% to 6.89% to 7.40%. Schools positive in both earlier waves are positive again 60.2% of the time, versus 4.2% for those never positive.

- **COVID check still open:** 2021-22 is itself a post-pandemic year, so changes from 2017-18 mix trend and pandemic effects. The 2020-21 file is needed to separate them.

Recommended target definition

- **Primary, `any_seclusion`:** seclusion_instances ⩾ 1. Prevalence 7.2% in the modeling frame.
- **Secondary, `elevated_seclusion`:** seclusion_instances ⩾ 5 and rate ⩾ 1 per 100 students. Prevalence 3.2% (2,830 schools).
- **Modeling frame:** analysis-ready, enrollment ⩾ 30 (drops 3,255 schools), excluding the 641 implausible-FTE schools, with cross-validation grouped by district. Keep rows with missing %IDEA, %504 or %EL and add indicators. For rate work add shrinkage (about 30-100 pseudo-students).
- **Not recommended:** raw rate or count as a regression target.

Candidate feature shortlist

- **Include:** log enrollment, grade band, special-education / alternative / charter / justice flags, %IDEA, %504, %EL (each with a missing indicator), psychologists and social workers per 100 students, log district enrollment.
- **Optional:** magnet flag, counselors, nurses and teachers per 100, security staff per 100 (reverse-causality caution).
- **Second model only:** state (or region).
- **Drop:** raw teacher and counselor FTE, `support_fte_per_100` (derivation unclear), `lea_schools` (0.97 with district enrollment), nonbinary count and partial flags.
- **Not usable:** virtual indicator (no outcome for virtual schools).
- **Exclude as leakage:** the seclusion component counts and rate. Restraint and other discipline measures are not in the file; physical restraint leaked strongly in 2015-16 (AUC 0.84 alone), so keep them out.

Data issues for the cleaning owners

- **High:** all 2,496 virtual schools have a reserved seclusion code and are dropped; confirm what -9 means for them.
- **High:** Section 504 and English-learner counts are 19-23% missing (-9), by state.

- **High:** the FTE outlier flag fires on 1 school, but 641 fail plausibility rules (for example law-enforcement FTE of 100 at 11 schools, up to 4,958 teachers per 100 students).
- **High:** extreme counts need verification (110 schools with more instances than students).
- **Medium:** nonbinary partial flags carry no information; 14 state-operated-program IDs in the analysis file with the keep/drop decision still open; 9 unresolved mapping rows and open definitions (preschool in totals, partial sums); race, sex and restraint columns not carried through.
- **Low:** two flags never fire; `seclusion_rate` is instances per 100 students; 2020-21 not supplied.

The full list with row counts is in `data_issues_202122.csv`.

Open items

1. Supply the 2020-21 cleaned file to complete the COVID check.
2. Carry race, sex and restraint columns through cleaning, then rerun the equity and leakage sections.
3. Team decision: minimum enrollment of 30, whether the headline model includes state, and how virtual schools and the 504 / English-learner gaps are handled.